

ECE 175: Computer Programming for Engineering
Applications Final Project: *The card game – Code Clash Uno*

Due Date: May 7th, 2025

ZyBooks 17.24 See late submission policy on the last page

Administrative details:

- The project is to be worked on in a team of two students.
Email uaece175@gmail.com your name and name of your team member by 11:59pm on Thursday April 24, 2025.
- **Honor code:** Your team is expected to submit your own code. You may ask others for advice, and in general discuss the project, but

WRITE YOUR OWN CODE!

If any part of the code submitted by different teams is identical, **ALL involved parties will receive zero credit on the entire project and one letter reduction in their final grade.**

This policy will be very aggressively enforced. ALL submitted code will be checked with a plagiarism detection tool (e.g., <http://theory.stanford.edu/~aiken/moss/>).

Game Rules:

- 1) There are two or more players.
- 2) Each player gets 7 cards to start with.
- 3) The game goes until either 1 person gets rid of all their cards or the deck runs out of cards. If the deck runs out of cards, the person with the fewest cards wins.
- 4) The players can only place cards on their turn.
- 5) If it's a player's turn and they don't have the required card, they draw from the deck once and then it switches to the other player's turn.
- 6) The regular cards are 0-9 in red, yellow, green, or blue variety. And there are two of each.
- 7) There are 4 special cards, AND, OR, NOT, and Reverse.
- 8) The way the AND card works is similar to a draw 4 card. The player that uses the AND card must pair it with a regular 0-9 card. The other player must place a card with the same color AND the same number. If the other player does not have a card that fits that description, they will draw 4.
- 9) The way the OR card works is also a variation of draw 4. The player that uses the OR card must pair it with a regular 0-9 card. The other player must place a card with the same color OR the same number. If the other player does not have a card that fits that description, they will draw 4.

10) The NOT card is placed by itself and will skip the other next player's turn. If there are only two players, the player that played the NOT card goes again.

11) The reverse card reverses the order of play. If there are only two players, the player that played the card will go again. If there are more than two players, the player that plays the card will go again and after that the order of play will be reversed.

12) Hand in a PDF or text file with instructions on how to play your game to facilitate grading and ensure your group gets the highest amount of points.

Project requirements:

Card game

- 1) To simulate the deck, and each of the players' hands, your team can use
- (a) arrays of struct card (each normal card consists of a color [red, yellow, green, and blue] and a number [0-9], special cards do not have a color

```
typedef struct card_s {  
    char name; //'0'-'9' for number cards, 'A' for AND, 'O' for OR, 'N' for  
    NOT, and 'R' for Reverse.  
    char color; //'R' for red, 'Y' for yellow, 'G' for green, 'B' for blue, and 'S'  
    for special cards  
} card;
```

OR

- (b) a dynamic list of cards [*Extra Credit* - represent your hand as a linked list of cards]

```
typedef struct card_s {  
    char name; //'0'-'9' for number cards, 'A' for AND, 'O' for OR, 'N'  
    for NOT, and 'R' for Reverse.  
    char color; //'R' for red, 'Y' for yellow, 'G' for green, 'B' for blue,  
    and 'S' for special cards  
  
    struct card_s *listp;  
  
} card;
```

If the dynamic list of cards is used to represent each of the players' hand, the cards removed from each of the players' hand are deleted from the list.

- 2) Create a full deck of 100 cards that are in order (two sets of 0-9 red cards, two sets of 0-9 yellow cards, two sets of 0-9 green cards, two sets of 0-9 blue cards, 5 AND cards, 5 OR cards, 5 NOT cards, and 5 Reverse cards).

Note 1: Your code must have a **CreateDeck()** function that creates a standard deck of 100 cards, that could be used for any card playing game.

- 3) You must then shuffle the deck, using the following algorithm:

- (a) For each card in the deck, get a random number in the range of 0 to 100 (to **shuffle** a deck of 100 cards) to be used as the index of the element to swap that

card with, i.e. if the first random number generated was 24, and the second random number generated was 98, the contents of index 24 (4 blue) and index 98 (Reverse card) will swap places. So now deck[24] is the Reverse card and deck[98] is the 4 blue card. Etc.

(b) Repeat step a) at least 10000 times.

Note 2: Your code must have a ***ShuffleCard()*** function using the algorithm explained above.

Note 3: It may be useful to add more functions such as deleteCard(), dealCard(), etc.

- 4) You must seed the random number generator with a call to time() with srand(). [see section 2.24 Random numbers in your ZyBooks]
- 5) After shuffling the deck, you must deal the cards by giving seven to each player.
- 6) After the cards are dealt to each player's hand, they should be displayed, i.e.

Player one's hand: //If "Player one" was entered by the first user.

Reverse
Blue 8
Blue 0
Green 5
Red 5
AND
Yellow 0

Player two's hand: //If "Player two" was entered by the second user.

OR
Red 0
Yellow 5
NOT
Green 9
AND
Yellow 6

- 7) Next play will begin. Player one gets to place a card on the pile first. The next player will then play a card. If Player one places a number card on the pile, Player 2 must place a card that has the same color or number. If they do not have a card with the same color or number, they will grab a card from the deck in their turn will be skipped. If Player 1 places a special card on the pile, the card will be in effect.

Note 4: The way your team displays information shown in item 6 above can be different, but it should convey the same information.

- 9) The game will continue until either one of the players runs out of cards or until the deck runs out of cards. If the deck runs out of cards, the player with the fewest cards wins.

- 10) *At the end of the game*, your code should
- announce the winner.
 - ask whether a user wants to play a new game.

Optional Features for extra credit:

1) (20 pts) Use dynamically allocated single-linked lists to simulate the deck of cards and each of the players' hands (see project requirement 1 under b) **OR** (10pts) use dynamically allocated struct arrays to simulate the deck of cards and each of the players' hands (see project requirement 1 under a). Choose only ONE. Doing both will not net you higher than 20 points extra credit.

2) (10 pts) Computer Players. Program a form of artificial intelligence for computer players.

Points distribution: See Rubric for final project grading.

Sample Execution Code Execution (User inputs are bold and colored red):

Enter number of players: **2**

Enter first player's name: **Player One**

Enter second player's name: **Player Two**

Player One's hand:

Reverse
Blue 8
Blue 0
Green 5
Red 5
AND
Yellow 0

Player Two's hand:

OR
Red 0
Yellow 5
NOT
Green 9
AND
Yellow 6

Card pile is empty.

Player One, enter which card to play from 0 to 6: **7**

Invalid choice, Player One does not have 8 cards

Player One, enter which card to play from 0 to 6: **2**

Player One's hand:

Reverse
Blue 8
Green 5
Red 5
AND
Yellow 0

Player two's hand:

OR
Red 0
Yellow 5
NOT
Green 9
AND
Yellow 6

Top of card pile is: Blue 0

Player Two, enter which card to play from 0 to 6: **4**

Invalid choice, cannot place Green 9 in Blue 0

Player Two, enter which card to play from 0 to 6: **1**

Player One's hand:

Reverse
Blue 8
Green 5
Red 5
AND
Yellow 0

Player two's hand:

OR
Yellow 5
NOT
Green 9
AND
Yellow 6

Top of card pile is: Red 0

Player One, enter which card to play from 0 to 5: **5**

Player One's hand:

Reverse
Blue 8
Green 5
Red 5
AND

Player two's hand:

OR
Yellow 5
NOT
Green 9
AND
Yellow 6

Top of card pile is: Yellow 0

Player Two, enter which card to play from 0 to 5: **1**

Player One's hand:

Reverse
Blue 8
Green 5
Red 5

AND

Player two's hand:

OR
NOT
Green 9
AND
Yellow 6

Top of card pile is: Yellow 5

Player One, enter which card to play from 0 to 4: **0**

Player One's hand:

Blue 8
Green 5
Red 5
AND

Player two's hand:

OR
NOT
Green 9
AND
Yellow 6

Top of card pile is: Reverse

Player One, enter which card to play from 0 to 3: **0**

Player One's hand:

Green 5
Red 5
AND

Player two's hand:

OR
NOT
Green 9
AND
Yellow 6

Top of card pile is: Blue 8

Player Two, enter which card to play from 0 to 4: **0**

Choose the card to play with OR from 1, 2, 3, 4: **1**

Cannot play two special cards in one turn

Choose the card to play with OR from 1, 2, 3, 4: **4**

Player One's hand:

Green 5
Red 5
AND

Player two's hand:

NOT
Green 9
AND

Top of card pile is: Yellow 6 with OR

Player One has no card that matches Yellow or 6

OR penalty, Draw 4

Player One's hand:

Green 5
Red 5
AND

Green 9
Yellow 5
NOT
Red 8

Player two's hand:

NOT
Green 9
AND

Top of card pile is: Yellow 6 with OR

Player Two, enter which card to play from 0 to 2: **2**

Choose the card to play with AND from 1, 2: **1**

Player One's hand:

Green 5
Red 5
AND
Green 9
Yellow 5
NOT
Red 8

Player two's hand:

NOT

Top of card pile is: Green 9 with AND

Player One, enter which card to play from 0 to 6: **3**

Card Matches, no AND penalty

Player One's hand:

Green 5
Red 5
AND
Yellow 5
NOT
Red 8

Player two's hand:

NOT

Top of card pile is: Green 9

Player Two, enter which card to play from 0 to 0: **0**

Next Player's turn skipped.

Player Two wins!

Play again? **N**

Goodbye